

# HEART DISEASE

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*Risk Prediction Using*

**Machine Learning**

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PRESENTED BY

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Course: Machine Learning 1 – BIA-5302

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# Introduction & Problem Statement



## Introduction



Leading cause of death worldwide



Often detected at **later stages**



Early prediction helps **prevent complications**



Machine learning **detects hidden patterns** in data



Supports **better and faster** medical decisions



## Problem Statement



Symptoms are often **unclear or not visible**



Traditional diagnosis **can be slow**



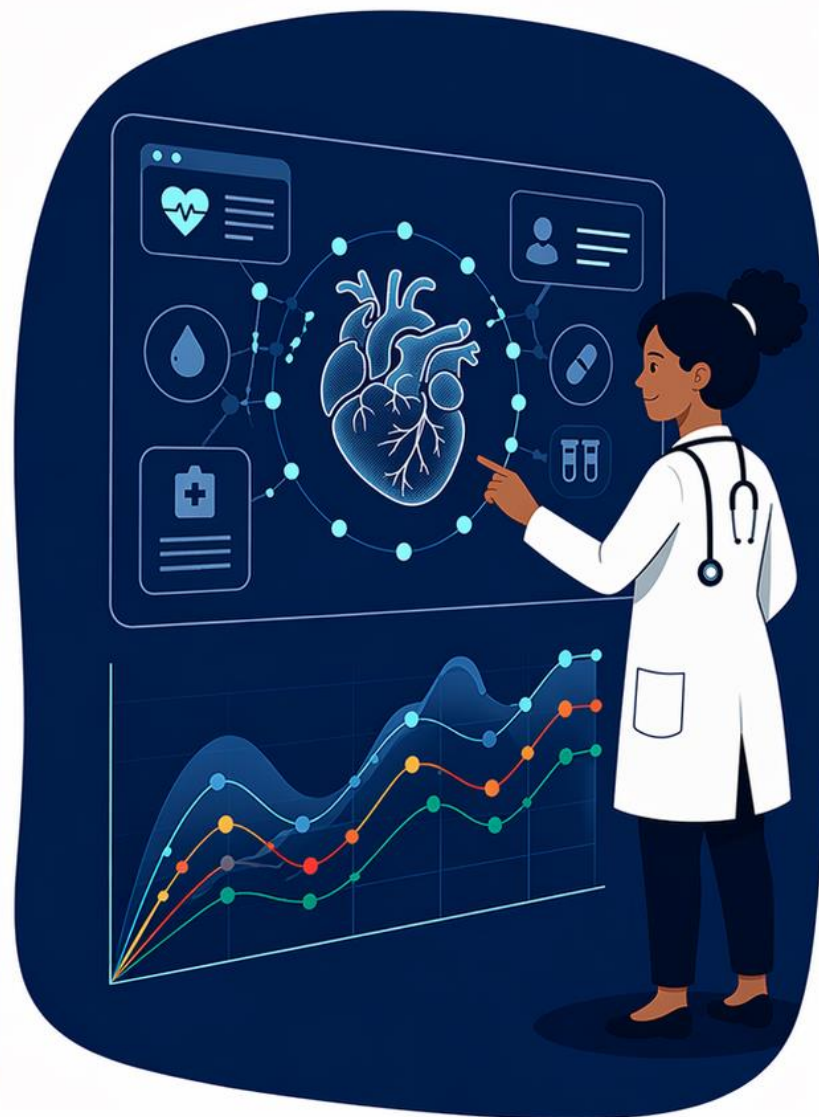
Difficult to analyze **large medical data** manually



May **miss important patterns**



Need for **faster and more accurate** prediction system





## Primary Goal:

Predict heart disease risk using medical data



## Compare Three Machine Learning Models

- Logistic Regression (Baseline Model)
- Decision Tree (Interpretable Model)
- K-Nearest Neighbors (K-NN) (Distance-Based Model)



## Evaluate Models Using Key Metrics

Evaluate model performance using **Accuracy**, **Precision**, **Recall**, and **F1 Score**



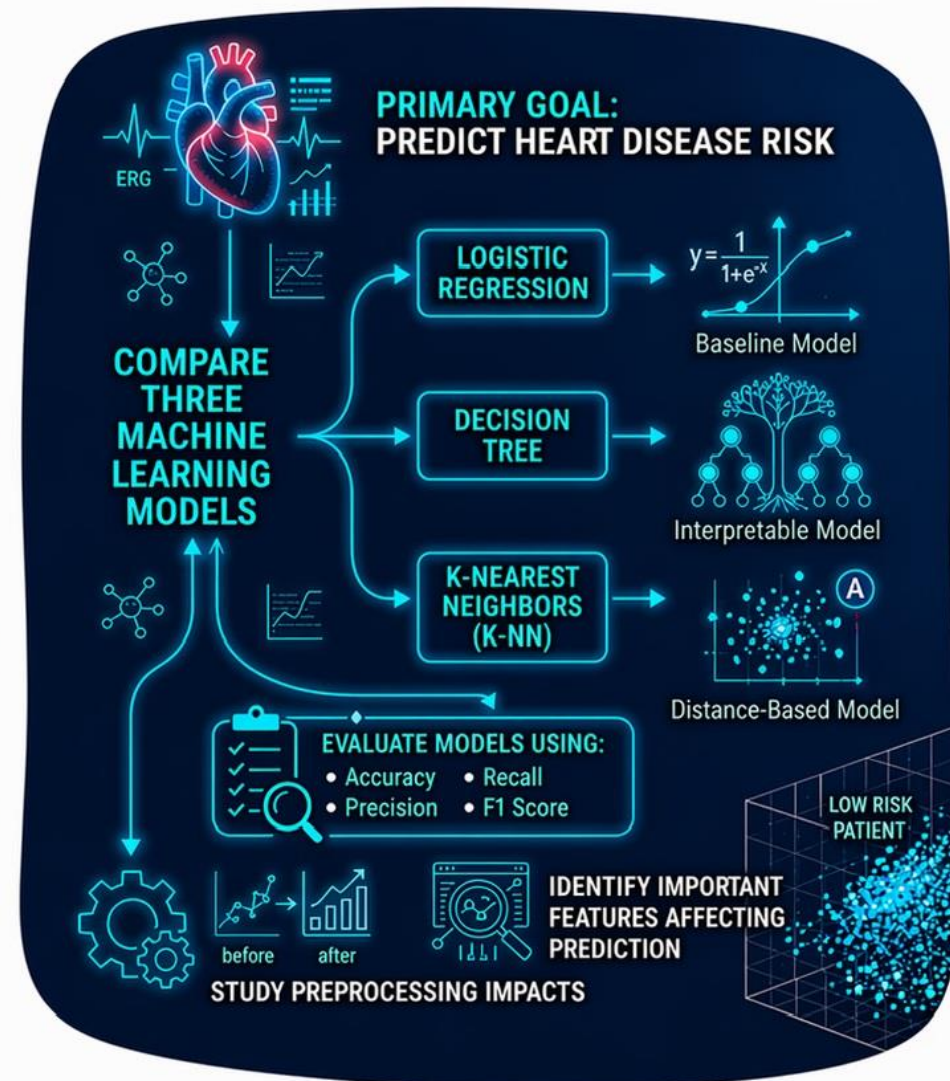
## Identify Important Features Affecting Prediction

Analyze and highlight the key factors that influence heart disease risk.

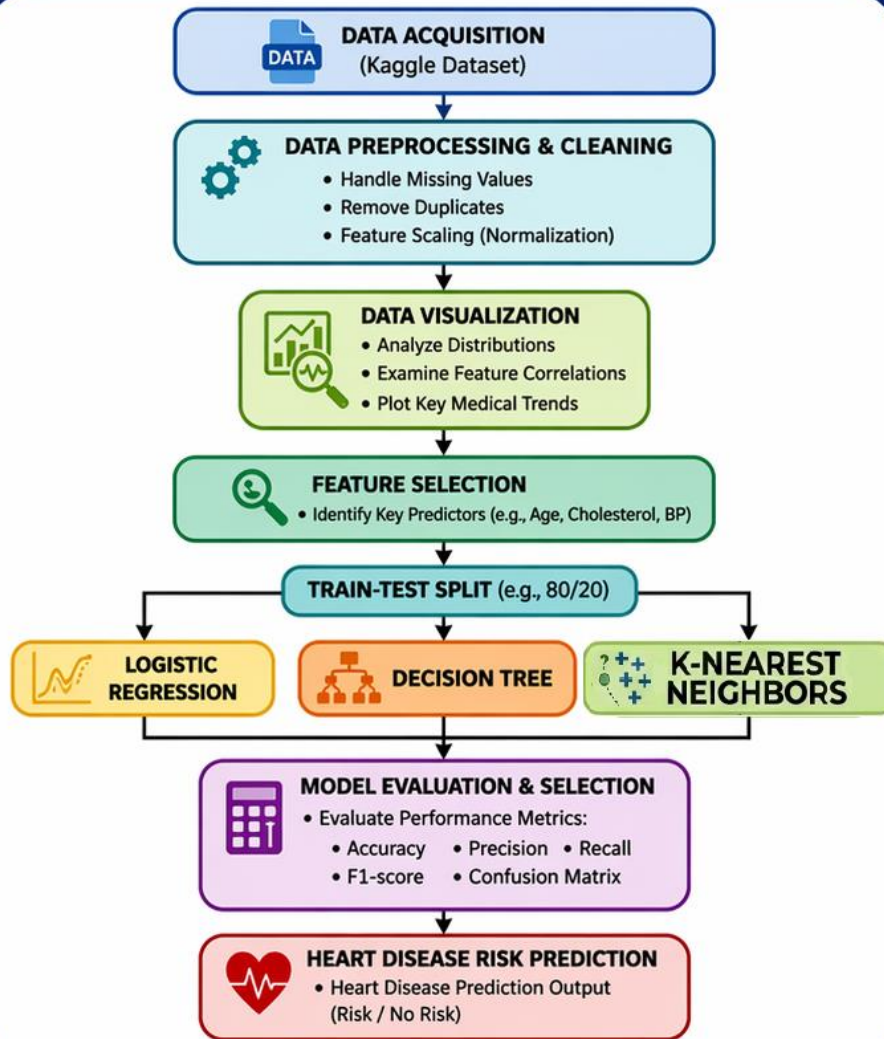


## Study How Preprocessing Impacts Model Performance

Evaluate how data cleaning, encoding, and scaling improve prediction accuracy.







- 01 Collect Dataset from Kaggle**  
Obtain the heart disease dataset from Kaggle for analysis.
- 02 Clean Data and Handle Missing Values**  
Remove duplicates and fix missing or inconsistent data.
- 03 Apply Preprocessing (Encoding and Scaling)**  
Convert categorical variables and scale numerical features.
- 04 Visualize Data to Understand Patterns**  
Use charts and graphs to explore distributions and correlations.
- 05 Split Data into Training and Testing Sets**  
Divide the data to train and evaluate the models fairly.
- 06 Train Models and Evaluate Performance**  
Train multiple models and compare their results using evaluation metrics.
- 07 Select Best Model for Prediction**  
Choose the best-performing model for accurate heart disease prediction.

# Dataset Description

## Data Source

- Kaggle – Heart Disease Diagnosis Dataset

## Original Source

UCI Machine Learning Repository

## Dataset Overview

- 270 patient records
- 14 columns (13 features + 1 target)

## Dataset Type

- Medical classification dataset

## Key Features

- Age
- Cholesterol
- Resting Blood Pressure
- Maximum Heart Rate
- Chest Pain Type

## Target Variable

Binary classification:

- 1= No Heart Disease
- 2= Heart Disease

Dataset Shape:  
(270, 14)

Dataset Info:  
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 270 entries, 0 to 269  
Data columns (total 14 columns):

# Data Cleaning & Preprocessing

**Missing Values** : No missing values were found in the dataset.

**Duplicate Removal** : No duplicate records were found.

**Column Standardization** : Cleaned column names for consistency.

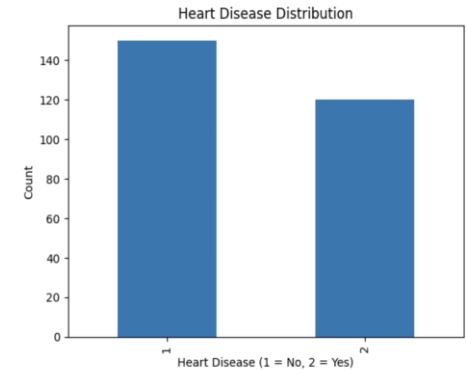
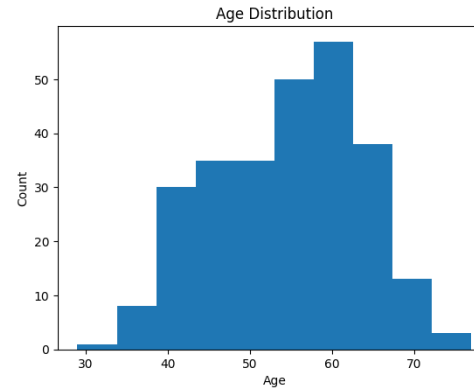
**Feature Scaling** : Applied StandardScaler.

**Train-Test Split** : 80% training / 20% testing.

# Data Visualization

## Age Distribution

Most patients between 40–65 years

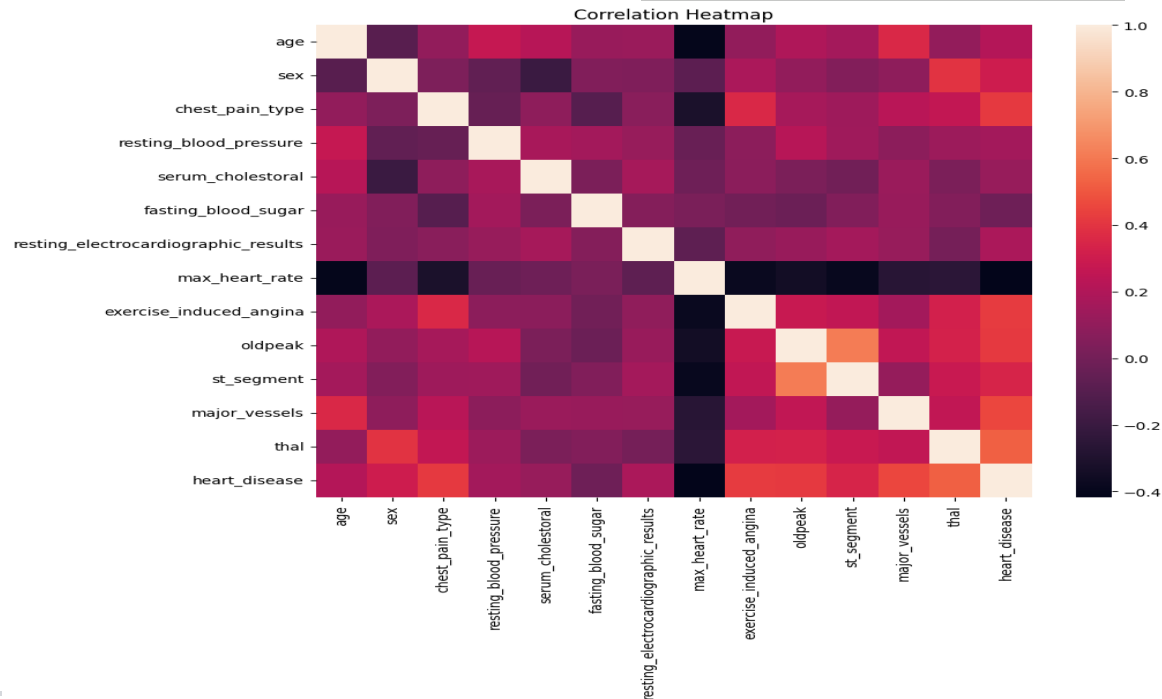


## Target Distribution

Shows number of patients with heart disease

## Correlation Heatmap

Shows relationships between variables



# Model 1 — Logistic regression

## Model explanation

- Predicts heart disease (yes/no) using patient features such as age, cholesterol, and blood pressure.
- Outputs probability between 0 and 1.
- Feature scaling applied before training.

## Training results

- Trained on 80% of the dataset, tested on 20%.
- High accuracy and stable performance.

Accuracy

**90.7%**

## Confusion matrix

|             | Pred: No | Pred: Yes |
|-------------|----------|-----------|
| Actual: No  | 31       | 2         |
| Actual: Yes | 3        | 18        |

TN = 31

TP = 18

FP = 2

FN = 3

*Only 5 errors out of 54 predictions*



## Model explanation

- Splits data into decision rules based on features.
- Easy to visualize and interpret.

## Training results

- Initial model overfitted the data (68.5% accuracy).
- Improved by limiting tree depth (max\_depth = 3).

Accuracy (after tuning)

**72.2%**

↑ from 68.5%

## Confusion matrix

|             | Pred: No | Pred: Yes |
|-------------|----------|-----------|
| Actual: No  | 26       | 7         |
| Actual: Yes | 8        | 13        |

TN = 26

TP = 13

FP = 7

FN = 8

*15 errors out of 54 predictions*

# Model 3 — K-Nearest Neighbors (KNN)

## Model explanation

- Classifies patients by finding the K most similar cases in training data and assigning the majority class outcome.
- Distance-based method requiring feature scaling — StandardScaler applied before training.

## K selection

- Tested K = 1 to 20 by comparing train vs. test accuracy.
- Optimal K = 12 selected — highest test accuracy without overfitting.

## Training results

- Trained on 80% of dataset (216 samples), tested on 20% (54 samples).

Accuracy

**90.7%**

## Confusion matrix

|             | Pred: No | Pred: Yes |
|-------------|----------|-----------|
| Actual: No  | 33       | 0         |
| Actual: Yes | 5        | 16        |

TN = 33

TP = 16

FP = 0

FN = 5

*0 false positives — perfect precision for no-disease class*

**Precision: 86.8% | Recall: 100% | F1-Score: 93.0%**  
*(for No Disease class)*

# Model Comparison

## Performance Summary — All Three Models

| Model               | Accuracy | Precision | Recall | F1-Score |
|---------------------|----------|-----------|--------|----------|
| Logistic Regression | 90.7%    | 91.2%     | 93.9%  | 92.5%    |
| Decision Tree       | 72.2%    | 76.5%     | 78.8%  | 77.6%    |
| KNN (K=12)          | 90.7%    | 86.8%     | 100%   | 93.0%    |

## Key Insights

- Logistic Regression & KNN both achieve 90.7% accuracy — tied for best overall performance.
- KNN achieves perfect 100% recall for No Disease class — zero false positives.
- Decision Tree scores lowest (72.2%) due to overfitting, despite being most interpretable.
- `thal`, `major_vessels`, and `chest_pain_type` are the most important features across all models.

## Recommended Model: Logistic Regression

*Best balance of accuracy, precision, recall, and interpretability for healthcare settings.*

# Results

## Best Performing Model

- Logistic Regression & KNN (K=12) tied at 90.74% accuracy — Decision Tree at 72.22%
- Outperformed others on precision, recall, and F1-score metrics

## Key Findings

- Chest pain type is the most predictive feature for heart disease risk
- Maximum heart rate achieved and ST depression are strong secondary predictors
- Feature scaling significantly improved Logistic Regression performance

## Observations

- Preprocessing quality had a measurable impact on all three models tested
- KNN achieved 100% precision for the disease class but missed 5 cases (76% recall)

# Model Results Summary

## Logistic Regression

Accuracy

**90.7%**

Confusion Matrix

|                 |                 |
|-----------------|-----------------|
| TN<br><b>31</b> | FP<br><b>2</b>  |
| FN<br><b>3</b>  | TP<br><b>18</b> |

*5 errors out of 54 predictions*

## Decision Tree

Accuracy

**72.2%**

Confusion Matrix

|                 |                 |
|-----------------|-----------------|
| TN<br><b>26</b> | FP<br><b>7</b>  |
| FN<br><b>8</b>  | TP<br><b>13</b> |

*15 errors out of 54 predictions*

## K-NN (K=12)

Accuracy

**90.7%**

Confusion Matrix

|                 |                 |
|-----------------|-----------------|
| TN<br><b>33</b> | FP<br><b>0</b>  |
| FN<br><b>5</b>  | TP<br><b>16</b> |

*5 errors out of 54 predictions*



# Results & Discussion

## Model Accuracy Comparison

Logistic Regression



K-NN (K=12)



Decision Tree



01

### Top predictive feature

Chest pain type was the strongest individual predictor of heart disease risk across all three models.

02

### Secondary predictors

Max heart rate achieved and ST depression were consistently strong secondary features in driving classification.

03

### Preprocessing impact

Feature scaling significantly improved both Logistic Regression and K-NN accuracy; cleaning and encoding boosted all models.

04

### Instance-based advantage

K-NN with K=12 tied Logistic Regression at 90.7% accuracy and achieved 100% precision on the disease class, showing that a well-tuned distance-based model can rival linear baselines.

# Conclusion & Future Work

## CONCLUSION

- All three ML models successfully classified heart disease risk with varying accuracy levels.
- Logistic Regression and K-NN (K=12) tied at 90.7% test accuracy — the best performance observed in this project.
- K-NN achieved 100% precision on the disease class (zero false positives), while Logistic Regression delivered the most balanced precision–recall trade-off.
- ML offers a scalable, data-driven supplement to traditional clinical diagnosis pathways.

## FUTURE WORK

- **Larger, diverse datasets**  
Expand beyond 270 records (Cleveland Clinic) to multi-source, globally representative clinical data.
- **Deep learning & ensembles**  
Apply gradient-boosted trees (XGBoost, LightGBM) and neural tabular models to push diagnostic accuracy further.
- **EHR integration**  
Connect models to electronic health records for real-time patient risk screening in clinical settings.
- **Class balance & recall**  
Use SMOTE or cost-sensitive thresholds to address the mild class imbalance and reduce K-NN's 5 false negatives.

# FINAL REPORT — PROJECT SUMMARY

## Problem

Predict heart disease risk from patient clinical data using supervised ML to support early diagnosis.

## Dataset

270 patient records, 13 clinical features, binary target (0 = No Disease / 1 = Disease) – Kaggle/UCI.

## Models Tested

Logistic Regression · Decision Tree · K-NN – trained on 80/20 train-test split.

## Best Result

Logistic Regression and K-NN: 90.7% accuracy.  
Decision Tree: 72.2%.

## Key Insight

Chest pain type is the strongest predictor. Preprocessing quality directly determines model performance.

## Recommendation

Ensemble models + larger diverse datasets + EHR integration for real-world clinical deployment.

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